Title: Effects of Copper and lead exposure on the Physiology of *Hemigrapsus nudus*.

Abstract:

Heavy metal contamination in marine ecosystems is an environmental concern because pollutants such as copper and lead can persist in coastal habitats and disrupt normal physiological processes in aquatic organisms. Crustaceans are useful model organisms for studying contaminant stress because they are commonly exposed to pollutants through seawater and sediments while relying on regulated osmoregulatory and immune systems (Capparelli et al., 2016). This study examined how copper and lead exposure affected the physiology of the intertidal shore crab *Hemigrapsus nudus*. We hypothesized that heavy metal exposure would increase physiological stress, resulting in altered hemocyte counts, increased righting time and disrupted osmolality compared to control crabs. Crabs were exposed to control, copper or lead treatments over three weeks and physiological responses were assessed using behavioral righting time, hemolymph osmolality and hemocyte counts. Lead exposure caused the strongest behavioral and osmoregulatory stress with average righting times reaching 9.55 seconds and osmolality increasing to 1017 mmol/kg by week 3. Copper exposure produced the strongest immune response with the highest hemocyte counts observed. These findings suggest that different heavy metals affect different physiological systems in *H. nudus*, highlighting the ecological risk of contamination in coastal marine environments.

Introduction:

Heavy metal pollution from runoff, industrial discharge, stormwater contamination and other anthropogenic activities is a major concern in marine ecosystems because metals persist and can disrupt essential biological processes, including respiration, ion transport, immune function and metabolism (Thurberg et al.,1973; Capparelli et al., 2016).

Crustaceans are useful for studying contaminant stress because they frequently exposed through sediment and seawater, while relying on tightly regulated osmoregulatory systems vulnerable to metal stress. (Amado et al., 2006; Amado et al., 2012)

Osmoregulation is critical for maintaining stable ion and water balance, and disruption of gill transport mechanisms that can increase energetic stress (Adeleke et al., 2021). Heavy metals may also trigger immune activation, making hemocyte counts useful indicators of physiological stress. Behavioral measures such as righting time provide a simple assessment of impaired physiological performance.

The objective of this study was to determine how copper and lead exposure affect the physiology of *H. nudus* by measuring behavioral performance, osmoregulatory conditions and immune response. We predicted that heavy metal exposure would increase righting time, elevate hemolymph osmolality and alter hemocyte counts relative to control crabs.

Methods:

Study Organism

The intertidal shore crab *Hemigrapsus nudus* was used as the model organism for this experiment. This species was selected because it is common in Pacific coastal environments and is naturally exposed to fluctuating environmental conditions, making it suitable for examining stress physiology.

Experimental Design

This study used a comparative exposure design to evaluate physiological responses to heavy metal contamination. Crabs were assigned to one of three treatment groups:

- Control group: No heavy metal added
- Lead group: 29.18 g
- Copper group: 30.22 g

Crabs remained under treatment conditions for three weeks. During week 2, three crabs died, with one mortality occurring in each treatment group, reducing the sample size for later measurements. Water changes were performed every other day in the copper treatment to maintain water quality.

Three physiological indicators were measured:

1. Righting time as behavioral measure of physiological stress.
2. Hemolymph osmolality as a measure of osmoregulatory disrupt.
3. Hemocyte count as an indicator of immune activation.

Righting Time Measurements

Behavioral stress was assessed using righting time. Individual crabs were placed upside down and the time required to return to an upright position was recorded in seconds. Longer righting times were interpreted as indicators of impaired physiological performance or toxic stress.

Osmolality Measurements

Hemolymph samples were collected to assess internal osmotic balance. Osmolality was measured in mmol/kg and used as an indicator of physiological stress related to disrupted ion regulation.

Hemocyte Counts

Hemolymph samples were examined microscopically to quantify hemocyte abundance. Total visible hemocytes were counted and used as an indicator of immune response.

Data Analysis

Data were analyzed descriptively using averages and graphical comparisons. Righting times were averaged across treatment groups by week. Osmolality data were compared using boxplots and hemocyte counts were summarized by treatment group. Due to small sample sizes and mortality, formal statistical analyses were not performed.

Results:

Immune Response: Hemocyte counts

No hemocytes were observed in the control during week 3 (Figure 1). Lead treated crabs showed a total of 3 hemocytes, while copper treated crabs showed a total of 18 hemocytes.

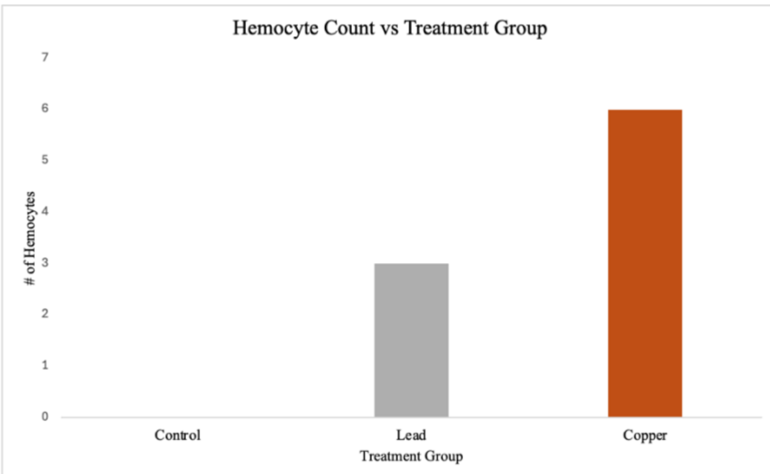

Figure 1: Bar graph showing hemocyte counts in *Hemigrapsus nudus* across different treatment groups (control, lead, copper).

Behavioral Responses: Righting Time

Lead exposure produced the strongest behavioral stress response (Figure 2). In week 1, average righting times were 1.89 seconds for control crabs, 2.65 seconds for copper treated crabs and 4.29 seconds for lead treated crabs.

By week 2, control crabs averaged 0.79 seconds, copper treated crabs averaged 1.74 seconds and lead treated crabs averaged 9.02 seconds.

In week 3, control crabs averaged 2.35 second, copper treated crabs averaged 1.28 seconds and lead treated crabs averaged 9.55 seconds.

Lead consistently cause the slowest recovery times, indicating sustained physiological stress.

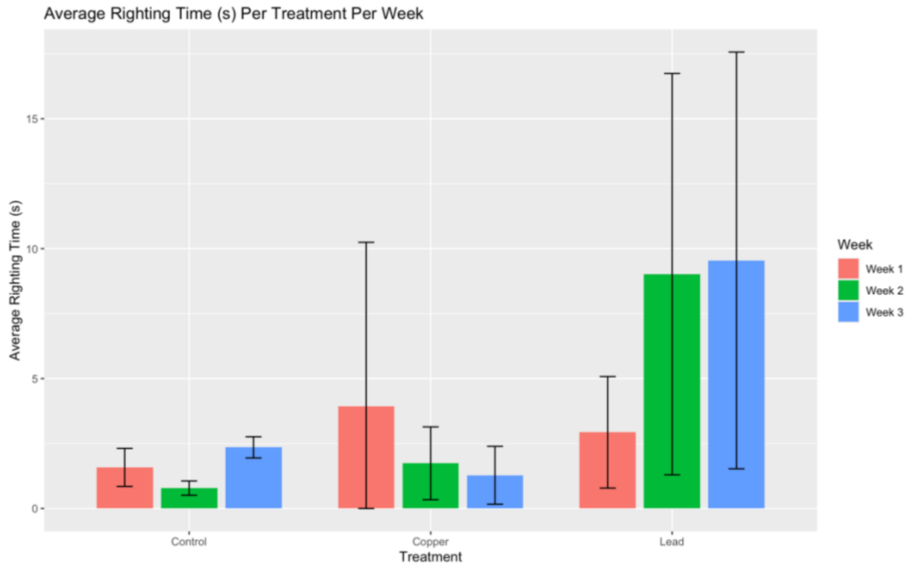

Figure 2. Average righting time (s) of *Hemigrapsus nudus* exposed to control, copper and lead treated crabs across three weeks.

Osmoregulatory Response: Osmolality

Osmolality values varied between treatment groups (Figure 3). In week 2, osmolality of the control group was 687 and 1035 mmol/kg, copper group was 736 and 877 mmol/kg and lead group was 800 and 778 mmol/kg.

By week 3, osmolality of the control group was 924 mmol/kg, copper group was 982 mmol/kg and lead group was 1017 mmol/kg.

Lead treated crabs showed the highest osmolality by week 3, suggesting greater osmoregulatory disruption.

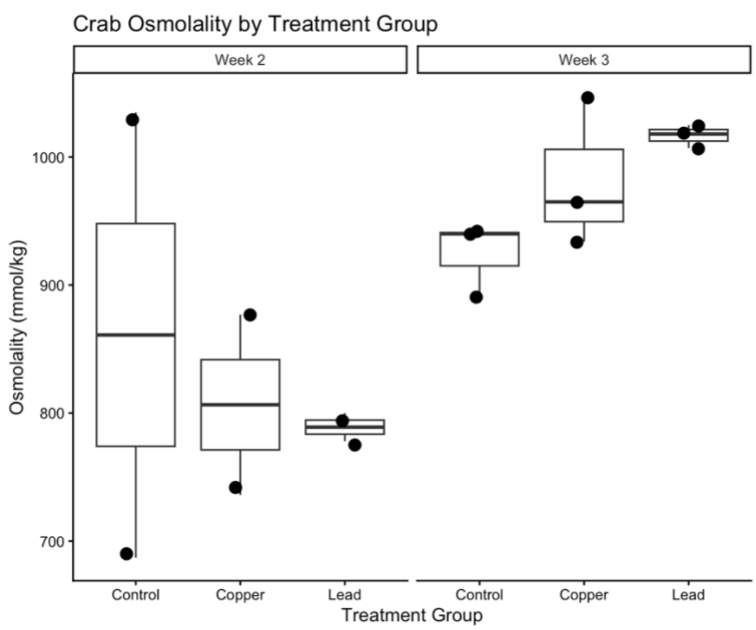

Figure 3. Hemolymph osmolality (mmol/kg) in *Hemigrapsus nudus* exposed to control, copper and lead treatments.

Discussions:

This study showed that copper and lead exposure affected multiple physiological systems in *Hemigrapsus nudus*, but the effects differed depending on the response measured.

Lead exposure produced the strongest behavioral impairment. Crabs exposed to lead showed longer righting times, specifically in week 2 and 3. Increased righting time suggests impaired movement, toxic stress or reduced physiological performance. Similar findings have been reported in previous crab studies, where heavy metal exposure disrupted energy metabolism and oxygen consumption (Thurberg et al., 1973; Capparelly et al., 2016)

Lead also caused the greatest osmoregulatory disruption. By week 3, lead treated crabs showed the highest hemolymph osmolality. This supports previous findings that lead interferes with ion regulation, specifically in gill tissues that are responsible for maintaining osmotic balance (Amado et al., 2006; Amado et al., 2012). Increased osmolality suggest that crabs were under physiological stress and likely spending additional energy to maintain homeostasis.

Copper exposure produced a different patter. While copper did not consistently impaired righting time, it triggered the strongest immune response, as shown by elevated hemocyte counts. This suggests copper may provoke a stronger immune activation or inflammatory stress than lead under these conditions.

These results are consistent with other studies showing that heavy metals can alter crab physiology in multiple ways, including disrupted osmoregulation and stress related metabolic changes (Adeleke et al., 2021; Capparelly et al., 2016).

One limitation of this study was that exposure was reported as total metal mass added rather than standardized concentration, making direct comparisons with published studies more difficult. Additionally, three crabs died during week 2, with one mortality in each treatment group, reducing sample size and increasing variability in later measurements. Another limitation was that water changes were performed every other day in the copper treatment, which may have changed copper concentration over time and contributed to the less consistent behavioral response observed in that group. Hemocyte counts were only measured at one time point, limiting interpretation of immune response trends over time.

Despite these limitations, the results suggest that heavy metal contamination can significantly affect physiological function in intertidal crabs.

Conclusion

Heavy metal exposure affected the physiology of *Hemigrapsus nudus* in measurable but different ways. Lead caused the strongest behavioral and osmoregulatory stress, while copper triggered the strongest immune response. These findings suggest that different contaminants may affect different physiological system. Future studies should use larger sample sizes, standardized concentrations and longer monitoring periods to better understand long term ecological consequences.

References:

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